RPA BUSINESS CASE

Customs Documentation Completeness Check Automation

Automation Anywhere RPA Implementation Proposal

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Platform	Automation Anywhere, CargoWise One ERP
Related Document	Customs Documentation Workflow, Gap Analysis BRD v1.0

This business case was developed following gap analysis findings documented in the Customs Documentation Workflow BRD. It proposes RPA automation as the implementation mechanism for the pre-submission validation checkpoint identified as Gap G2 in that analysis. All data is based on replicated scenarios from real freight forwarding operations and has been anonymised for portfolio use.

1. Executive Summary

This business case proposes the deployment of an RPA bot using Automation Anywhere to automate the customs documentation completeness check within the CargoWise One freight forwarding workflow. The automation directly addresses Gap G2 identified in the Customs Documentation Workflow Gap Analysis BRD, which found that no pre-submission validation checkpoint existed to catch documentation errors before they reached brokers and customs authorities.

The current manual process requires an operations staff member to individually retrieve each shipment record, check document completeness from memory, and assemble a documentation package before submission. This process is inconsistent, time-consuming, and responsible for approximately 12 percent of shipments experiencing documentation-related errors that result in clearance delays of 2.4 days on average.

The proposed RPA solution automates the completeness check end to end. The bot monitors CargoWise One for new booking confirmations, extracts shipment and document data, runs a rule-based completeness check against a predefined document matrix, logs results directly in CargoWise One, and only escalates to the operations team when an exception is identified. Based on current shipment volumes and time studies, the solution is projected to deliver a return on investment within seven months of go-live.

Metric	Current State	Post-Automation Target
Manual check time per shipment	Approximately 18 minutes	Less than 1 minute (exception handling only)
Documentation error rate	Approximately 12%	Less than 3%
Shipments requiring rework	Approximately 35%	Less than 5%
Estimated weekly manual effort	Approximately 18 hours	Less than 2 hours
Projected ROI timeline	Not applicable	7 months post go-live

2. Problem Statement

2.1 Current Process Pain Points

The customs documentation completeness check is performed manually by operations staff for every shipment processed through the freight forwarding operation. Based on the gap analysis conducted in June 2025 and documented in the related BRD, the following pain points were identified and quantified.

The check relies entirely on individual operator knowledge of required document sets, which varies by shipment type, origin, destination, and commodity. There is no system-enforced checklist and no validation step within CargoWise One that prevents an incomplete documentation package from being submitted externally. As a result, approximately 12 percent of shipments are submitted with missing or incorrect documentation, triggering broker rejections, customs holds, and rework cycles that average 2.4 days of additional clearance time per affected shipment.

The manual check also consumes significant operational time. Time studies conducted during the gap analysis estimated an average of 18 minutes per shipment for a complete check, inclusive of retrieving the record, reviewing attached documents, cross-checking against shipment details, and assembling the outgoing package. Across the shipment volumes processed, this equates to approximately 18 hours of operational staff time per week dedicated to a process that is fundamentally rule-based and repeatable.

2.2 Why RPA is the Right Solution

The customs documentation completeness check meets all four criteria that make a process well-suited for RPA automation. The process is highly rule-based and follows a consistent logic that can be codified without exception handling complexity. It is high volume and repetitive, executing identically for every shipment regardless of operator. It operates within a structured digital environment, CargoWise One, which provides accessible data through its interface and API layer. And it is time-sensitive, as delays in the check directly delay submission and clearance.

An RPA solution implemented using Automation Anywhere can replicate the human check steps end to end, applying the document matrix rules consistently across 100 percent of shipments without variation, fatigue, or oversight gaps. The bot executes in seconds rather than minutes, logs every check outcome directly in CargoWise One for auditability, and only surfaces exceptions to the operations team when a genuine gap is identified.

3. Proposed RPA Solution

3.1 Solution Overview

The proposed solution is an Automation Anywhere attended or unattended bot deployed within the existing CargoWise One environment. The bot operates continuously during business hours, monitoring for new shipment booking confirmations and executing the completeness check workflow automatically for each new booking without requiring any operator trigger.

The bot executes the following sequence for every shipment. It monitors CargoWise One for new booking confirmations triggered by the milestone event. It extracts the shipment record data including shipment type, origin, destination, commodity, and incoterms. It identifies the required document set for that shipment profile from the document matrix rules configured within the bot. It checks each required document against the attachments on the CargoWise One shipment record. If all required documents are present and the shipment details are consistent, it logs a validation pass against the record and triggers the pre-submission checkpoint clearance. If any document is missing or a discrepancy is detected, it generates an exception task assigned to the responsible operator with a two-hour resolution SLA, consistent with Requirement FR-006 from the Gap Analysis BRD.

3.2 Process Maps

The current state and future state process maps are provided as separate files in this repository. The current state map documents the manual process as it was operating prior to the proposed automation, including the identified pain points and rework loops. The future state map shows the redesigned process with RPA automation touchpoints marked, demonstrating where the bot replaces manual steps and where the operations team retains involvement for exception handling and final submission.

Refer to process_map_current_state.svg and process_map_future_state.svg in this repository.

3.3 Document Matrix — Bot Validation Rules

The bot applies a predefined document matrix to determine the required document set for each shipment. The matrix is configurable and maintained by the system administrator, allowing updates without redeployment of the bot logic.

Shipment Type	Required Documents	Additional Conditions
Import, General Cargo	Commercial invoice, packing list, bill of lading, customs entry form	Country of origin certificate required if preferential tariff claimed
Import, Dangerous Goods	Commercial invoice, packing list, bill of lading, customs entry form, dangerous goods declaration, MSDS	Packaging certificate required for certain UN classes
Export, General Cargo	Commercial invoice, packing list, bill of lading, export declaration	Certificate of origin required if destination country requires it
Export, Controlled Goods	Commercial invoice, packing list, bill of lading, export declaration, export permit	End user certificate required where applicable
Transshipment	Commercial invoice, packing list, master bill of lading, house bill of lading, transit declaration	Additional port authority documents where required by routing

4. Cost Benefit Analysis

4.1 Implementation Costs

Cost Item	Description	Estimated Cost
RPA Bot Development	Automation Anywhere bot design, development, testing, and deployment by RPA developer	AUD 12,000
UAT and Go-Live Support	Business Analyst and System Administrator time for UAT coordination, defect resolution, and go-live support	AUD 3,500
Automation Anywhere Licence	Unattended bot licence for production environment, annual cost	AUD 8,000 per year
CargoWise Integration Review	System Administrator review of CargoWise One data access configuration for bot	AUD 1,500
Training and Change Management	Operations team briefing and updated SOP documentation	AUD 800
Total Year 1 Cost		AUD 25,800
Annual Recurring Cost (Year 2 onwards)	Licence renewal and maintenance	AUD 9,500 per year

4.2 Projected Benefits

Benefit Area	Basis of Calculation	Estimated Annual Value
Operational time saving	Reduction from 18 hours to 2 hours per week of manual check time. 16 hours per week at AUD 35 per hour average operations staff rate.	AUD 29,120
Rework cost reduction	Reduction from 35 percent to 5 percent of shipments requiring rework. Estimated 1.5 hours rework per shipment at AUD 35 per hour, across estimated shipment volumes.	AUD 18,200
Clearance delay cost avoidance	Reduction of 2.4 day average delay on affected shipments. Storage and demurrage cost avoidance estimated at AUD 180 per day per shipment.	AUD 14,400
Client retention value	Reduced complaint rate from documentation errors, estimated to protect approximately 3 percent of annual revenue at risk from repeat errors.	AUD 22,000
Total Estimated Annual Benefit		AUD 83,720

4.3 Return on Investment

ROI Metric	Value
Total Year 1 Investment	AUD 25,800
Total Year 1 Benefit	AUD 83,720
Net Year 1 Benefit	AUD 57,920
ROI (Year 1)	224 percent
Payback Period	Approximately 3.7 months
3-Year Net Benefit (after all costs)	AUD 202,660

5. Risk Assessment

#	Risk	Description	Likelihood	Impact	Mitigation
R1	Document matrix incomplete	Bot flags false positives if the document matrix does not cover all shipment profile combinations	**Medium**	**High**	Comprehensive matrix review conducted before bot deployment. Operations team signs off on matrix completeness.
R2	CargoWise One data access changes	System updates to CargoWise One could affect bot data access paths and cause failures	**Low**	**High**	Bot built with resilience checks. System Administrator notified before any CargoWise One updates.
R3	Operator resistance to automation	Operations staff may be reluctant to trust bot validation outputs initially	**Medium**	**Medium**	Change management briefing included in implementation plan. Parallel run period planned for first two weeks post go-live.
R4	Bot handling edge cases incorrectly	Unusual shipment profiles not covered by matrix rules may be misclassified	**Low**	**Medium**	All unrecognised shipment profiles escalated to operator by default. No silent failures permitted.
R5	Licence cost increase	Automation Anywhere licence costs may increase at renewal	**Low**	**Low**	Annual licence cost reviewed against benefit realisation at each renewal. Alternative platforms assessed if cost exceeds threshold.

6. Implementation Approach

6.1 Implementation Phases

Phase	Activity	Owner	Duration
Phase 1, Discovery	Document matrix finalisation and CargoWise One data mapping. RPA developer briefing and bot design sign-off.	BA with System Administrator	1 week
Phase 2, Development	Automation Anywhere bot development covering monitoring, data extraction, rule application, logging, and exception escalation.	RPA Developer	2 weeks
Phase 3, Testing	Bot testing in CargoWise One UAT environment across all shipment types in the document matrix. Defect resolution.	RPA Developer with BA	1 week
Phase 4, Parallel Run	Bot runs alongside manual process for two weeks. Results compared to validate accuracy before manual process is retired.	Operations Team with BA	2 weeks
Phase 5, Go-Live	Manual completeness check retired. Bot operates as primary validation mechanism. Operations team handles exceptions only.	System Administrator	1 day
Phase 6, Review	30-day post go-live review of bot accuracy, exception rate, and time savings against projected benefits.	BA	Ongoing

6.2 Success Criteria

The implementation will be considered successful when the following criteria are met at the 30-day post go-live review.

1. Bot validation accuracy is greater than 97 percent across all shipment types processed
2. Documentation error rate has reduced from 12 percent to less than 3 percent of shipments
3. Manual completeness check effort has reduced from 18 hours to less than 2 hours per week
4. Zero silent bot failures — all unrecognised profiles are correctly escalated to the operations team
5. Operator satisfaction with the exception handling workflow is rated satisfactory or above in post-go-live feedback

7. Recommendation and Decision

Based on the gap analysis findings from the Customs Documentation Workflow BRD, the cost benefit analysis in this document, and the risk assessment above, I recommend proceeding with the Automation Anywhere RPA implementation as proposed.

The process is a strong automation candidate, the financial case is compelling with a 3.7-month payback period and 224 percent first-year ROI, and the risk profile is manageable with the mitigations outlined. The parallel run approach in Phase 4 provides a low-risk transition path that maintains operational continuity while confidence in the bot is established.

Approval is sought to proceed to Phase 1 Discovery and to commission the RPA developer engagement.

Decision	Approver	Date	Signature
Proceed with RPA implementation as proposed	Operations Manager		
Budget approval for Year 1 investment of AUD 25,800	Finance Director		

8. Document Control

Version	Date	Author	Changes
0.1	June 2025	N M Ziyaf	Initial draft following BRD gap analysis completion
0.2	June 2025	N M Ziyaf	Cost benefit analysis and risk assessment added
1.0	June 2025	N M Ziyaf	Final version with all sections complete